Master Study Guide: Rotational Dynamics & Equilibrium

Grade 11 Physics | Term 2 Topic 1

1. Classification of Motion

Before solving equations, identify the system's behavior:

- **Pure Translation:** Every point moves with the same $\vec{v}$ and $\vec{a}$. Line segments remain parallel to themselves (e.g., an elevator cabin).
- **Pure Rotation:** All points move in circles around a fixed axis (Δ). They share the same angular velocity (θ') and angular acceleration (θ''), but linear speed $v = r\theta'$ increases with distance from the center.
- **Combined Motion:** The object translates while rotating (e.g., a rolling wheel).

2. The Moment of a Force

The **Moment ($\mathcal{M}$)** is the rotational effect of a force. It depends on the magnitude of F, the point of application, and the direction.

The Scalar Formula

$$\mathcal{M}_{F/\Delta} = \pm F \cdot d_{\perp}$$

Where $d_{\perp}$ (the lever arm) is the perpendicular distance from the axis Δ to the force's line of action.

The Zero Moment Conditions

A force produces **zero** moment if:

1. The force is **parallel** to the axis of rotation.
2. The line of action of the force **intersects** the axis of rotation ($d_{\perp} = 0$).

3. Rotational Equilibrium

For a rigid body to remain in rotational equilibrium (static or constant θ'):

$$\sum \mathcal{M}_{ext/\Delta} = 0$$

- **Sign Convention:** Counter-clockwise (CCW) is the positive (+) direction; Clockwise (CW) is negative (-).
- **Pivot Choice:** You can choose any point as the axis, but choosing a point where an unknown reaction force acts (like a hinge) simplifies the calculation.

4. Moment of Inertia (I)

I represents the resistance of a body to changes in its rotational motion. It depends on mass distribution relative to the axis.

Key Formulas

- Point Mass: $I = mr^2$
- System of Particles: $I_{total} = \sum m_i r_i^2$
- Solid Disk/Cylinder: $I = \frac{1}{2}MR^2$
- Thin Hoop: $I = MR^2$
- Thin Rod (Pivot at end): $I = \frac{1}{3}ML^2$

5. Newton's Second Law for Rotation

The fundamental law governing varied rotational motion relates the resultant moment to the angular acceleration:

$$\sum \mathcal{M}_\Delta = I \cdot \theta''$$

Where:

- $\sum \mathcal{M}_\Delta$ is the sum of external moments in $N \cdot m$.
- I is the moment of inertia in $kg \cdot m^2$.
- θ'' is the angular acceleration in rad/s^2.

💡 Technical Pro-Tip (The "Crane Arm" Problem)

When analyzing a falling beam (like the Rotating Crane Arm exercise):

1. **Weight Torque:** The weight acts at the Center of Mass (G). If the rod is horizontal, $\mathcal{M}_W = -Mg \cdot \frac{L}{2}$.
2. **Instantaneous Acceleration:** The moment the support cable is cut, the initial angular acceleration is:

$$\theta'' = \frac{\mathcal{M}_{net}}{I_{total}}$$

3. **Linear Link:** The linear acceleration of the tip (point B) is $a_B = L \cdot \theta''$.

⚠️ Common Pitfalls to Avoid

- **Units:** Moment is measured in **Newton-meters (N·m)**.
- **Lever Arm:** Always ensure $d_\perp$ is the *perpendicular* distance. If the force is applied at an angle ϕ to the rod, $d_\perp = r \sin(\phi)$.
- **Algebraic Signs:** Be rigorous with the sign convention. A falling rod usually has a negative moment (CW), leading to a negative θ'' if CCW is taken as positive.